

Software Engineer AI Task

AI Developer Candidate Assignments

This document presents a practical assignment intended to assess software developers on their ability to address real-world AI-enabled business challenges. The assignment focuses on a distinct business problem and offers developers an opportunity to demonstrate their technical and problem-solving skills. Candidates are permitted & encouraged to use any AI technology of their choosing but ultimately, they are accountable for the code and solution. They are required to host their solution, share the source code, and provide clear documentation detailing their work.

The use of Generative AI tools is permitted and encouraged; however, any such usage must be clearly and transparently cited in the submission.

Deliverables for Assignment

- Publicly accessible hosted URL for the application
- Source code repository (e.g., GitHub or equivalent)
- README or documentation that includes:
 - Problem statement
 - Solution approach
 - Technology stack used
 - Instructions for building and running the project locally
 - Details of AI tools, models, and references used
 - Design decisions and trade-offs made during development
- Use any LLM of their choice (use API keys and state their references)
- Any other useful information which will help us understand/assess the assignment

Evaluation Rubric – AI Developer Take-Home Assignments

Dimension	Evaluation Criteria
Problem Understanding & Solution Fit	Clearly defines the problem and delivers a well-scoped, stakeholder-aligned solution.
AI Usage & Reasoning	Uses AI intentionally with appropriate models, prompts, and awareness of limitations.
System Architecture & Design	Implements a clean, modular architecture aligned to the problem scope.
Code Quality & Engineering Practices	Produces readable, maintainable code with sensible abstractions and error handling.
Output Quality & UX Clarity	Delivers clear, structured outputs through an intuitive and trustworthy UX.
Reliability & Edge Case Handling	Handles invalid inputs and failures gracefully without crashing.
Documentation & Communication	Provides clear setup instructions, design explanation, and AI tool citations.

Market Research Intelligence Assistant

Business Problem

Product and GTM teams struggle to stay current on **competitor activity and market trends** because relevant information is scattered across blogs, websites, announcements, and articles. Build a lightweight application that helps a user **collect, analyze, and summarize market intelligence** from multiple public sources.

Assignment Description

Create a web application that allows a user to:

1. Provide a **list of competitor names or topics**
2. Provide **one or more source URLs** (blogs, announcement pages, articles)
3. Generate a **structured market intelligence summary**, including:
 - Key themes or trends
 - Notable competitor activities
 - Source references for each insight
 - Check for hallucination using LLM-as-a-judge to verify the claims in the output report

Expected Scope (Baseline)

- A user interface for entering research intent (such as competitors, topics, source URLs, and optional context)
- Implement a research phase on the provided user input
- AI-generated summary grouped by themes
- Source traceability (which insight came from which URL)
- Implement authentication
- Simple persistence (store previous runs or links to analyze)
- Deploy the application using a cloud platform (Azure preferred)

Stretch Goals (Optional)

- Change detection (“what’s new since last run”)
- Simple monitoring

Success Criteria

- The output is **useful, structured, and readable**
- Insights are **clearly grounded in sources**
- The system handles **at least 3–5 URLs reliably**
- The app can be accessed via a **public URL**

Expected Deliverables

- Live hosted URL
- Source code repository (GitHub or equivalent)
- README with:
 - Problem statement
 - Architecture & tech stack
 - Local build & run instructions
 - AI tools / models / libraries used (with citations)
- Written explanation of design decisions